Assignment 3 - AutoML

Athlete Total-Lift Prediction with AutoGluon, MLflow, and H2O AutoML

Course	ADSP 31021 Machine Learning Operations
Author	Anuradha Rakh
Primary platform	AutoGluon with MLflow
Platform mode	full-code
Repository	https://github.com/anuradharakh/mlops_automl

Executive Summary

This project evaluates AutoGluon with MLflow and H2O AutoML using identical cleaned train, validation, and test partitions. Each platform is run with all approved features and its top three features.

Metric	Result
Recommended run	AutoGluon - all_features
Best model	WeightedEnsemble_L2
Validation RMSE	125.783
Test RMSE	124.679
Test MAE	91.376
Test R-squared	0.796

Key conclusion

The final recommendation is selected primarily by validation RMSE and confirmed using the untouched test set. Speed, feature count, interpretability, and deployment complexity are secondary selection factors.

1. Dataset and Reproducible Design

Item	Value	Item	Value
Raw rows	423,006	Final rows	53,505
Rows removed	29	Approved features	13
Train rows	34,243	Validation rows	8,561
Test rows	10,701	Target range	8 to 2,330
Random seed	42	Primary metric	Validation RMSE

Quality and leakage controls

- The target is the sum of deadlift, clean and jerk, snatch, and back squat.
- Target components, total_lift, and athlete identifiers are excluded from predictors.
- Sentinel lift value 1 is treated as missing before target construction.
- Twenty-nine corrupted target rows are audited and removed rather than capped.
- Features above 70% missingness are excluded before modeling.
- The same fixed 64/16/20 partitions are used across all current experiments.

Workflow overview

Stage	Purpose
1	Profile and validate the athlete dataset
2	Build leakage-safe features and fixed data partitions
3	Run AutoGluon with all features and top three features
4	Repeat the workflow with H2O AutoML
5	Train conventional baselines and compare platforms
6	Generate reports, validate evidence, and package submission

2. AutoGluon Primary Workflow

AutoGluon is classified as full-code AutoML. It automates model training, preprocessing, algorithm selection, hyperparameter exploration, leaderboard ranking, and ensembling where supported. Python and YAML still control data quality, feature eligibility, validation design, runtime budgets, tracking, and final approval.

Feature set	Count	Best model	Val. RMSE	Test RMSE	Test R2
All	13	WeightedEnsemble_L2	125.783	124.679	0.796
Top 3	3	WeightedEnsemble_L2	142.568	141.974	0.735

Top three by validation RMSE - all features

Rank	Model	Validation RMSE	Speed
1	WeightedEnsemble_L2	125.783	0.010
2	CatBoost	125.943	4.526
3	ExtraTreesMSE	131.008	1.483

Top three by speed - all features

Rank	Model	Validation RMSE	Speed
1	WeightedEnsemble_L2	125.783	0.010
3	ExtraTreesMSE	131.008	1.483
4	RandomForestMSE	132.286	4.455

Top five AutoGluon features: **gender, fran, weight_height_ratio, grace, pullups**.

Top-three-feature run: **gender, fran, weight_height_ratio**.

3. H2O AutoML and Baseline Context

Feature set	Count	Best model	Val. RMSE	Test RMSE	Test R2
All	13	GBM_2_AutoML_1_20260731_161446	130.707	129.437	0.780
Top 3	3	GBM_1_AutoML_1_20260731_162003	143.579	142.488	0.733

Top three by validation RMSE - all features

Rank	Model	Validation RMSE	Speed
1	GBM_2_AutoML_1_20260731_161446	130.707	716.000
2	GBM_3_AutoML_1_20260731_161446	130.723	840.000
3	GBM_4_AutoML_1_20260731_161446	131.010	1,556.000

Top three by training speed - all features

Rank	Model	Validation RMSE	Speed
14	GBM_grid_1_AutoML_1_20260731_161446_mod...	132.375	599.000
8	GBM_5_AutoML_1_20260731_161446	131.565	618.000
20	GLM_1_AutoML_1_20260731_161446	277.233	627.000

Top five H2O features: **gender, weight_height_ratio, fran, weight, grace**.

Assignment 1 historical baseline

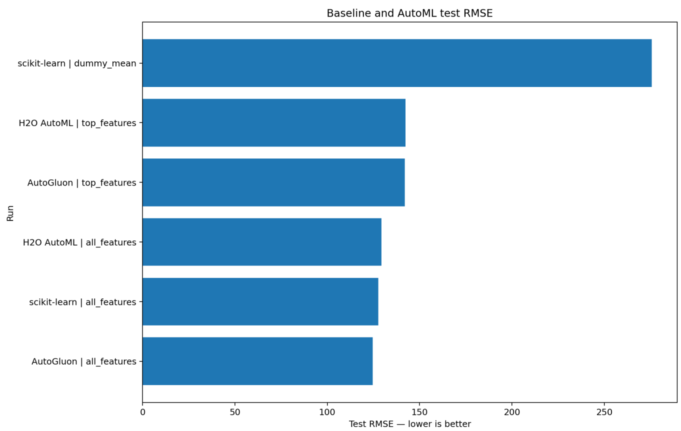
Model	Dataset	Split	Test RMSE	Test MAE	Test R2
RandomForestRegressor	Dataset v2 - processed	80 percent train / 20 percent tes	152.552	114.887	0.706

Assignment 1 used a different processed dataset and 80/20 split, and did not retain separate validation or reliable runtime evidence. Its metrics are historical context. The Phase 4 same-split Random Forest is the controlled current-workflow baseline.

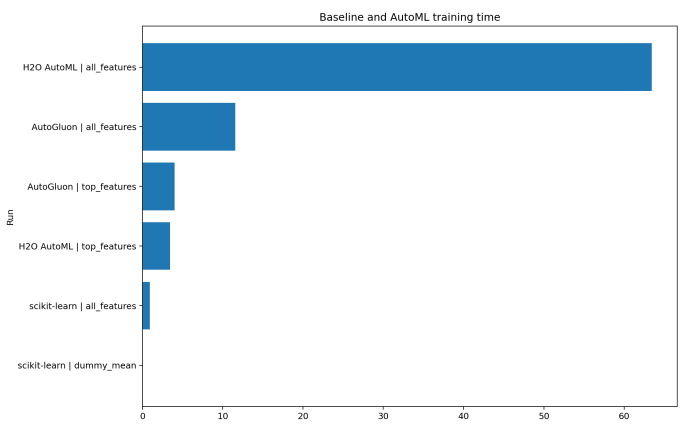
4. Cross-Platform Comparison and Recommendation

Platform	Features	Count	Model	Val. RMSE	Test RMSE	Test R2	Train sec.
AutoGluon	all_features	13	WeightedEnsemble_L2	125.783	124.679	0.796	11.573
scikit-learn	all_features	13	RandomForestRegressor	128.306	127.665	0.786	0.911
H2O AutoML	all_features	13	GBM_2_AutoML_1_2026073...	130.707	129.437	0.780	63.457
AutoGluon	top_features	3	WeightedEnsemble_L2	142.568	141.974	0.735	3.990
H2O AutoML	top_features	3	GBM_1_AutoML_1_2026073...	143.579	142.488	0.733	3.420
scikit-learn	dummy_mean	0	DummyRegressorMean	277.282	275.787	-0.000	0.001

Test RMSE comparison



Training-time comparison



Recommendation

The strongest run by validation RMSE is **AutoGluon - all_features**, model **WeightedEnsemble_L2**. It achieved test RMSE **124.679**, test MAE **91.376**, and test R-squared **0.796**.

AutoGluon and H2O shared 4 of their top five features. Feature importance is model-dependent and does not imply causality.

Operational implications

- AutoML reduces manual model selection and tuning effort.

- Ensembles can improve accuracy while increasing deployment complexity.
- Reduced-feature models simplify scoring and monitoring when validation quality is maintained.
- Runtime comparisons depend on hardware and platform measurement definitions.
- Production use still requires schema, quality, drift, latency, and cost monitoring.

Reproducibility

The repository includes Python 3.11 dependency files, YAML configurations, fixed random seeds, deterministic partitions, automated experiment scripts, MLflow tracking, saved leaderboards, tests, a submission validator, and a clean ZIP builder. Run **python scripts/finalize_submission.py** after all experiments pass.